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Equations for shear 3PCF

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1. OVERVIEW

Our code implements a calculation of the eight shear three-point functions given a set of cosmological parameters. The computation can be understood as divided into three steps, for which we will present the modeling equations.

- Computation of the matter bispectrum.
- Computation of the convergence bispectrum.
- Integration of the shear 3PCF.

2. MATTER BISPECTRUM

For the Bihalofit computation of the bispectrum, we calculate the following from CLASS:

- The linear matter power spectrum $P(k, z)$ on a grid of k and z values.
- The non-linear scale k_{nl} as for each redshift value.
- The comoving distance $\chi(z)$ and the derivative $dz/d\chi(z)$.

Next, we compute the fitting parameters, as functions of $\log \sigma_8$ and of $n_{\rm eff}$. We divide the parameters into global and dependent parameters, the former being k-independent, and the latter k-dependent. The global parameters are:

$$\begin{aligned}
\log b_n &= -3.428 - 2.681 \log \sigma_8 + 1.624 \log {\sigma_8}^2 - 0.095 \log {\sigma_8}^3 \\
\log c_n &= 0.159 - 1.107 n_{\rm eff} \\
\log \gamma_n &= 0.182 + 0.57 n_{\rm eff} \\
\log f_n &= -10.533 - 16.838 n_{\rm eff} - 9.3048 n_{\rm eff}^2 - 1.8263 n_{\rm eff}^3 \\
\log g_n &= 2.787 + 2.405 n_{\rm eff} + 0.4577 n_{\rm eff}^2 \\
\log h_n &= -1.118 - 0.394 n_{\rm eff} \\
\log m_n &= -2.605 - 2.434 \log \sigma_8 + 5.710 \log {\sigma_8}^2 \\
\log n_n &= -4.468 - 3.080 \log \sigma_8 + 1.035 \log {\sigma_8}^2 \\
\log \mu_n &= 15.312 + 22.977 n_{\rm eff} + 10.9579 n_{\rm eff}^2 + 1.6586 n_{\rm eff}^3 \\
\log \nu_n &= 1.347 + 1.246 n_{\rm eff} + 0.4525 n_{\rm eff}^2 \\
\log p_n &= 0.071 - 0.433 n_{\rm eff} \\
\log d_n &= -0.483 + 0.892 \log \sigma_8 - 0.086 \Omega_m \\
\log e_n &= -0.632 + 0.646 n_{\rm eff}
\end{aligned} \tag{1}$$

In order to compute the dependent parameters, we sort each set of (k_1, k_2, k_3) to find the set $(k_{textmin}, k_{\rm mid}, k_{\rm max})$. Then, we define $r_1 \equiv k_{\rm min}/k_{\rm max}$ and $r_2 = (k_{\rm mid} + k_{\rm min} - k{\rm max})/k_{\rm max}$. Now we compute the dependent parameters:

$$\begin{aligned}
\log a_n &= -2.167 - 2.944 \log \sigma_8 - 1.106 \log {\sigma_8}^2 - 2.865 \log {\sigma_8}^3 - 0.310 r_1 \gamma_n \\
\log \alpha_n &= min(-4.348 - 3.006 n_{\rm eff} - 0.5745 n_{\rm eff}^2 + 10^{-0.9+0.2 n_{\rm eff}} r_2^2, n_s) \\
\log \beta_n &= -1.731 - 2.845 n_{\rm eff} - 1.4995 n_{\rm eff}^2 - 0.2811 n_{\rm eff}^3 + 0.007 r_2
\end{aligned} \tag{2}$$

From these parameters, we compute the one-halo term:

$$B_{1h}(k_1,k_2,k_3,z)=\prod_{i=1,2,3}\frac{1}{a_nq_i^{\alpha_n}+b_nq_i^{\beta_n}}\frac{1}{1+(c_nq_i)^{-1}} \tag{3}$$

where $q_i = k_i/k_{nl}$.

The 2-halo and 3-halo effects are put together in the bihalofit model B_{3h} term. To compute this part, we take the F_2 kernel defined as

$$F_2(k_1,k_2,k_3)=\frac{5}{7}+\frac{2}{7k_1^2k_2^2}\left(\frac{-k_3^2+k_1^2+k_2^2}{2}\right)^2-\frac{-k_3^2+k_1^2+k_2^2}{2}\left(\frac{1}{2k_1^2}+\frac{1}{2k_2^2}\right) \tag{4}$$

Next, we compute the enhanced power spectrum:

$$P_{\text{enhanced}}(k_i,z)=\frac{1+f_nq_i^2}{1+g_nq_i+h_nq_i^2}P_L(k_i,z)+\frac{1}{m_nq_i^{\mu_n}+n_nq_i^{\nu_n}}\frac{1}{1+(p_nq_i)^{-3}} \tag{5}$$

Finally, we get

$$B_{3h}(k_1,k_2,k_3,z)=2(F_2(k_1,k_2,k_3)+d_nq_3)\prod_{i=1,2,3}\frac{1}{1+e_nq_i}P_{\text{enhanced}}(k_1,z)P_{\text{enhanced}}(k_2,z)+\text{two permutations} \tag{6}$$

3. CONVERGENCE BISPECTRUM

We now compute the projected convergence bispectrum $B_\kappa(\ell_1,\ell_2,\ell_3)$. From the Limber approximation, we take $k_i \approx \ell_i/\chi(z)$. We load the selection function $n(z)$ of the source galaxies and write the lensing kernel as:

$$g(\chi)=\int_\chi^{\chi_{\max}} d\chi' n(z(\chi'))\frac{\chi'-\chi}{\chi'} \tag{7}$$

The convergence bispectrum is now found by

$$B_\kappa(\ell_1,\ell_2,\ell_3)=\frac{27}{8}\left(\frac{H_0}{c}\right)^6\Omega_m^3\int_0^{\chi_{\max}} d\chi\frac{1}{\chi}(1+z(\chi))^3g(\chi)^3B(\ell_1/\chi,\ell_2/\chi,\ell_3/\chi,z(\chi)) \tag{8}$$

4. COSMIC SHEAR THREE POINT CORRELATION FUNCTIONS

The natural shear components Γ^0, Γ^1, Γ^2 and Γ_3, with the projection relative to the orthocenter, are computed by

$$\begin{aligned}\Gamma^0(x_1,x_2,x_3)=\frac{2\pi}{3}\int_0^\infty\frac{d\ell_1\ell_1}{(2\pi)^2}\int_0^\infty\frac{d\ell_2\ell_2}{(2\pi)^2}\int_0^{2\pi}d\phi B_\kappa(\ell_1,\ell_2,\sqrt{\ell_1^2+\ell_2^2+2\ell_1\ell_2\cos\phi})\\ \times e^{21\beta}((e^{i(\phi_1-\phi_2-6\alpha_3)})J_6(A_3)+(e^{i(\phi_3-\phi_2-6\alpha_1)})J_6(A_1)+(e^{i(\phi_3-\phi_1-6\alpha_2)})J_6(A_1))\end{aligned} \tag{9}$$

$$\begin{aligned}\Gamma^1(x_1,x_2,x_3)=\frac{2\pi}{3}\int_0^\infty\frac{d\ell_1\ell_1}{(2\pi)^2}\int_0^\infty\frac{d\ell_2\ell_2}{(2\pi)^2}\int_0^{2\pi}d\phi B_\kappa(\ell_1,\ell_2,\sqrt{\ell_1^2+\ell_2^2+2\ell_1\ell_2\cos\phi})\\ \times(e^{i(\phi_1-\phi_2+2\phi_3)})e^{2i(\beta-\phi-\alpha_3)}J_2(A_3)+(e^{i(\phi_3-\phi_2)})e^{-2i(\beta+\alpha_1)}J_2(A_1)+(e^{i(\phi_3-\phi_1-2\phi_2)})e^{2i(\beta+\phi-\alpha_2)}J_2(A_1)\end{aligned} \tag{10}$$

$$\Gamma^2(x_1,x_2,x_3)=\Gamma^1(x_2,x_3,x_1) \tag{11}$$

$$\Gamma^3(x_1,x_2,x_3)=\Gamma^1(x_3,x_1,x_2) \tag{12}$$

where the values for ϕ_i, β, A_i and α_i are given by:

$$
\begin{aligned}
\cos\phi_1 &= -\frac{x_1^2 - x_2^2 - x_3^2}{2x_2x_3} \\
\cos\phi_2 &= -\frac{x_2^2 - x_3^2 - x_1^2}{2x_3x_1} \\
\cos\phi_3 &= -\frac{x_3^2 - x_1^2 - x_2^2}{2x_1x_2} \\
\cos 2\beta &= \frac{(\ell_1^2 + \ell_2^2)\cos\phi + 2\ell_1\ell_2}{\ell_1^2 + \ell_2^2 + 2\ell_1\ell_2\cos\phi} \\
\sin 2\beta &= \frac{(\ell_1^2 - \ell_2^2)\sin\phi}{\ell_1^2 + \ell_2^2 + 2\ell_1\ell_2\cos\phi} \\
A_1 &= \sqrt{(\ell_2x_3)^2 + (\ell_3x_2)^2 + 2\ell_2x_2\ell_3x_3\cos\phi + \phi_1} \\
A_2 &= \sqrt{(\ell_1x_3)^2 + (\ell_3x_1)^2 + 2\ell_1x_1\ell_3x_3\cos\phi + \phi_2} \\
A_3 &= \sqrt{(\ell_1x_2)^2 + (\ell_2x_1)^2 + 2\ell_1x_1\ell_2x_2\cos\phi + \phi_3}
\end{aligned} \tag{13}
$$

$$
\begin{aligned}
\cos(\alpha_1) &= \frac{(\ell_2x3 + \ell_3x_2)\cos\frac{\phi+\phi_1}{2}}{A_1} \\
\sin(\alpha_1) &= \frac{(\ell_2x3 - \ell_3x_2)\sin\frac{\phi+\phi_1}{2}}{A_1} \\
\cos(\alpha_2) &= \frac{(\ell_3x1 + \ell_1x_3)\cos\frac{\phi+\phi_2}{2}}{A_2} \\
\sin(\alpha_2) &= \frac{(\ell_3x1 - \ell_1x_3)\sin\frac{\phi+\phi_2}{2}}{A_2} \\
\cos(\alpha_3) &= \frac{(\ell_1x2 + \ell_2x_1)\cos\frac{\phi+\phi_3}{2}}{A_3} \\
\sin(\alpha_3) &= \frac{(\ell_1x2 - \ell_2x_1)\sin\frac{\phi+\phi_3}{2}}{A_3}
\end{aligned}
$$

The integrals can be rewritten in terms of new variables ψ and R. We write:

$$\psi = \arctan\frac{\ell_2}{\ell_1} \tag{14}$$

$$R = \sqrt{\ell_1^2 + \ell_2^2} \tag{15}$$

and find

$$
\begin{aligned}
\Gamma^0(x_1,x_2,x_3) &= \frac{1}{6(2\pi)^3}\int_0^{2\pi} d\phi \int_0^{\pi/2} d\psi \sin 2\psi \int_0^\infty dR e^{2i\beta} J_6(R) T^0(x_1,x_2,x_3,\phi,\psi,R) \\
\Gamma^1(x_1,x_2,x_3) &= \frac{1}{6(2\pi)^3}\int_0^{2\pi} d\phi \int_0^{\pi/2} d\psi \sin 2\psi \int_0^\infty dR J_2(R) T^1(x_1,x_2,x_3,\phi,\psi,R)
\end{aligned} \tag{16}
$$

where

$$
\begin{aligned}
T^0(x_1,x_2,x_3,\phi,\psi,R) = &\frac{e^{i(\phi_2-\phi_3-6\alpha_1)}}{A_1'^4} B_\kappa\left(\frac{R}{A_1'}\cos\psi, \frac{R}{A_1'}\sin\psi, \frac{R}{A_1'}\sqrt{1+2\cos\psi\sin\psi\cos\phi}\right) \\
&+\frac{e^{i(\phi_3-\phi_1-6\alpha_2)}}{A_2'^4} B_\kappa\left(\frac{R}{A_2'}\cos\psi, \frac{R}{A_2'}\sin\psi, \frac{R}{A_2'}\sqrt{1+2\cos\psi\sin\psi\cos\phi}\right) \\
&+\frac{e^{i(\phi_1-\phi_2-6\alpha_3)}}{A_3'^4} B_\kappa\left(\frac{R}{A_3'}\cos\psi, \frac{R}{A_3'}\sin\psi, \frac{R}{A_3'}\sqrt{1+2\cos\psi\sin\psi\cos\phi}\right)
\end{aligned} \tag{17}
$$

and

$$
\begin{aligned}
T^1(x_1,x_2,x_3,\phi,\psi,R) = {} & \frac{e^{i(\phi_3-\phi_2-2\alpha_1-2\beta)}}{A_1'^4} B_\kappa\left(\frac{R}{A_1'}\cos\psi, \frac{R}{A_1'}\sin\psi, \frac{R}{A_1'}\sqrt{1+2\cos\psi\sin\psi\cos\phi}\right) \\
& + \frac{e^{i(2\phi-2\alpha_2+2\beta+\phi_3-\phi_1-2\phi_2}}{A_2'^4} B_\kappa\left(\frac{R}{A_2'}\cos\psi, \frac{R}{A_2'}\sin\psi, \frac{R}{A_2'}\sqrt{1+2\cos\psi\sin\psi\cos\phi}\right) \\
& + \frac{e^{i(-2\phi-2\alpha_3+2\beta+\phi_1-\phi_2+2\phi_3)}}{A_3'^4} B_\kappa\left(\frac{R}{A_3'}\cos\psi, \frac{R}{A_3'}\sin\psi, \frac{R}{A_3'}\sqrt{1+2\cos\psi\sin\psi\cos\phi}\right)
\end{aligned} \tag{18}
$$

with $A_i' = A_i/R$.

5. COORDINATE TRANSFORMATION

Or computed result for the Γ^i functions now can be transformed from orthocenter coordinates to centroid. The length of the line that bisects x_3 and goes to the vertez opposite x_3 is:

$$h_3 = \frac{\sqrt{2x_1^2+2x_2^2-x_3^2}}{2} \tag{19}$$

The angle between this line and side x_3 is given by ψ_3. We have:

$$\cos 2\psi_3 = \frac{(x_2^2-x_1^2)^2 - 4x_1^2x_2^2\sin^2\phi_3}{4h_3^3x_3^3} \tag{20}$$

and

$$\sin 2\psi_3 = \frac{(x_2^2-x_1^2)x_1x_2\sin\phi_3}{h_3^3x_3^3} \tag{21}$$

which gives us

$$\exp 2i\psi_3 = \frac{(x_2^2-x_1^2)^2 - 4x_1^2x_2^2\sin^2\phi_3}{4h_3^3x_3^3} + i\frac{(x_2^2-x_1^2)x_1x_2\sin\phi_3}{h_3^3x_3^3} \tag{22}$$

The other ψ_i angles are defined accordingly. Our new Γ^i functions can now be written as

$$\Gamma_C^0 = \Gamma_O^0 \exp 2i(\psi_1+\psi_2+\psi_3) \tag{23}$$

$$\Gamma_C^1 = \Gamma_O^1 \exp 2i(-\psi_1+\psi_2+\psi_3) \tag{24}$$

$$\Gamma_C^2 = \Gamma_O^2 \exp 2i(\psi_1-\psi_2+\psi_3) \tag{25}$$

$$\Gamma_C^3 = \Gamma_O^3 \exp 2i(\psi_1+\psi_2-\psi_3) \tag{26}$$

6. PLOTS

REFERENCES

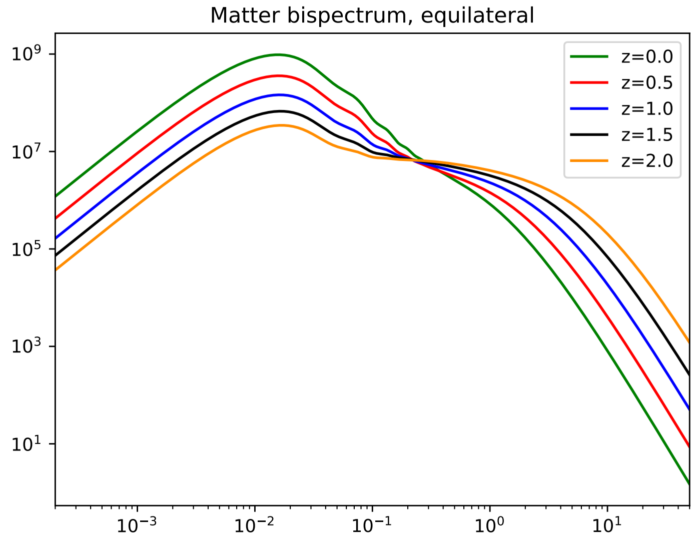

Figure 1. Equilateral matter bispectra (Bihalofit)

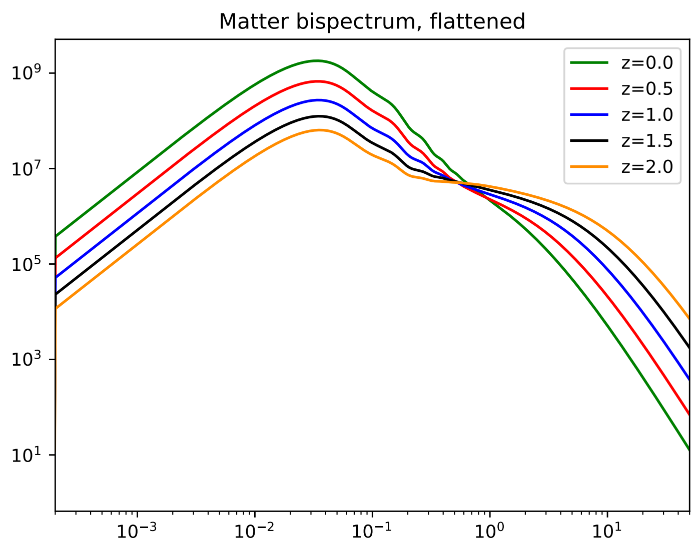

Figure 2. Flattened matter bispectra (Bihalofit)

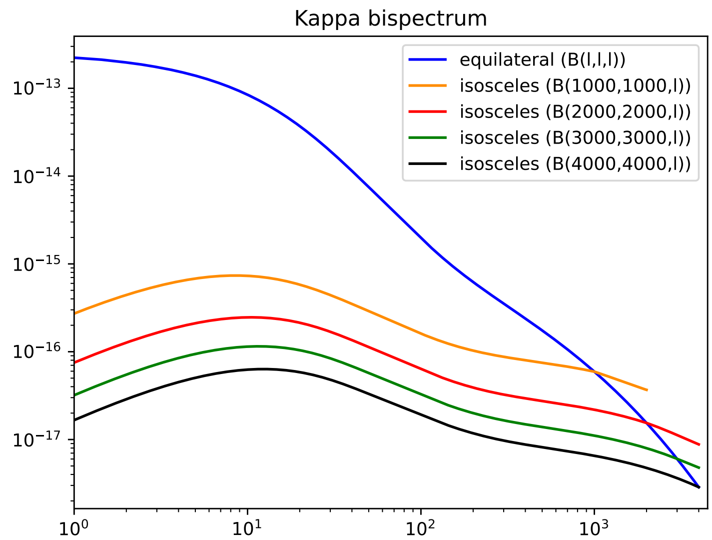

Figure 3. Convergence bispectra (Bihalofit)

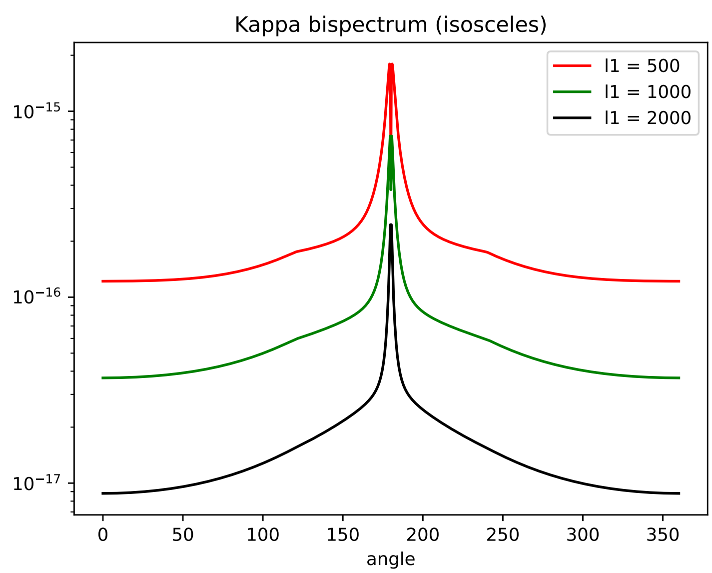

Figure 4. Convergence bispectra per angle (Bihalofit)

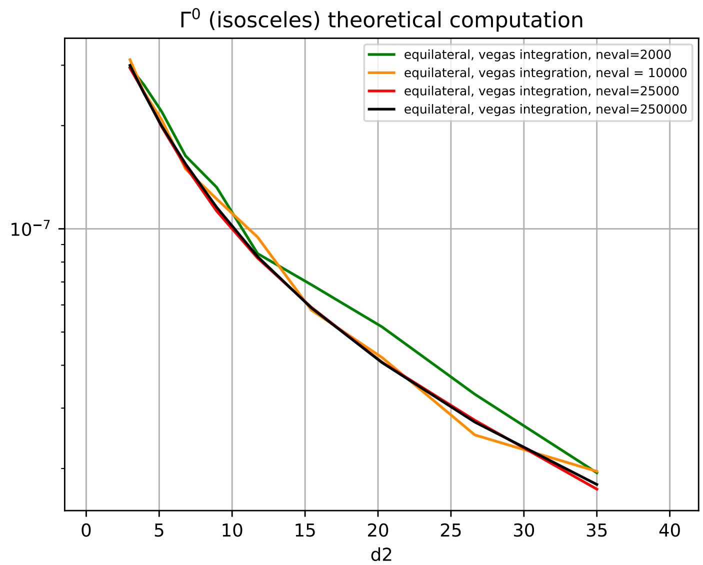

Figure 5. (Not yet updated) Isosceles Γ^0 shear 3pt function for a fiducial cosmology, with different number of evaluations. In all cases, 5 iterations were taken. Computation time is approximately 0.0002 seconds per d_2 value per evaluation.

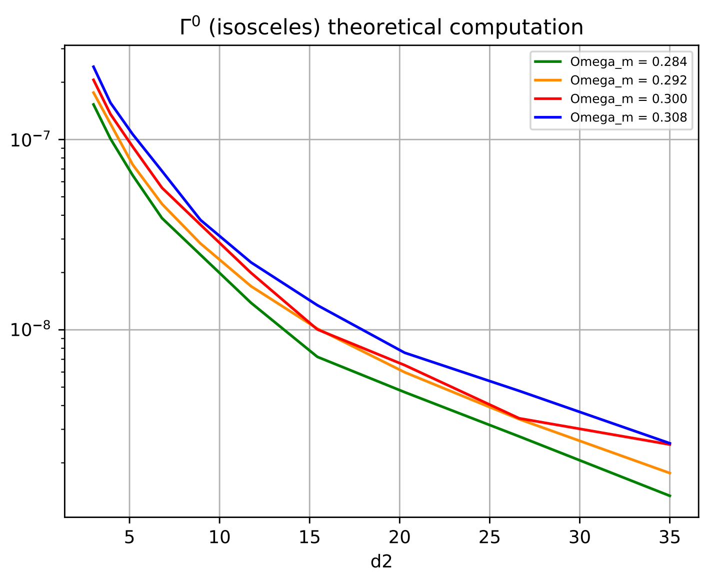

Figure 6. Isosceles Γ^0 shear 3pt function for varying Ω_m. Number of evaluations: 60000. Number of iterations: 8

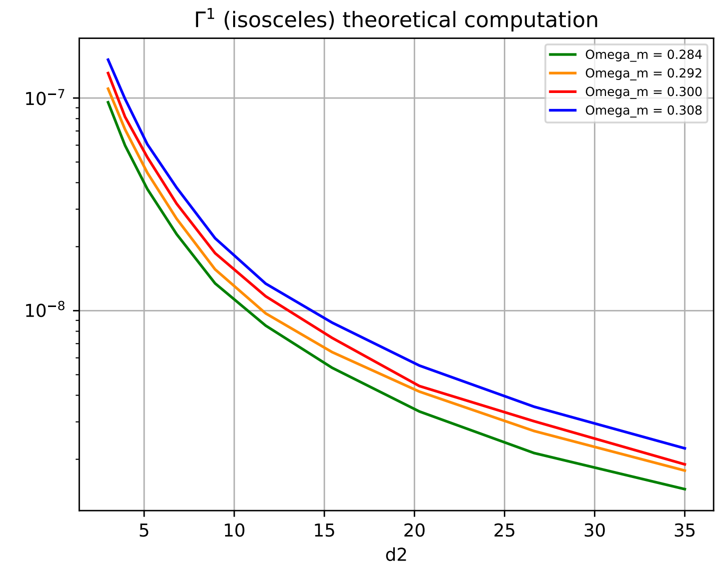

Figure 7. Isosceles Γ^1 shear 3pt function for varying Ω_m. Number of evaluations: 60000. Number of iterations: 8

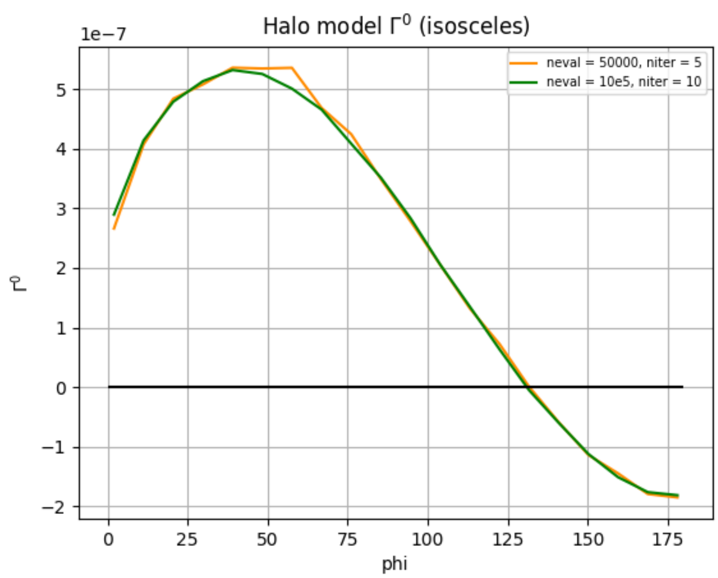

Figure 8. Isosceles Γ^0 shear 3pt function as a function of opening angle ϕ. For the smooth result we used 10 iterations with 10e5 evaluations each for each one of 20 ϕ values. Total time for Γ^0 is $0.0002s \times 10e5 \times 10 \times 20 \times 2 \approx 160\text{min}$.)

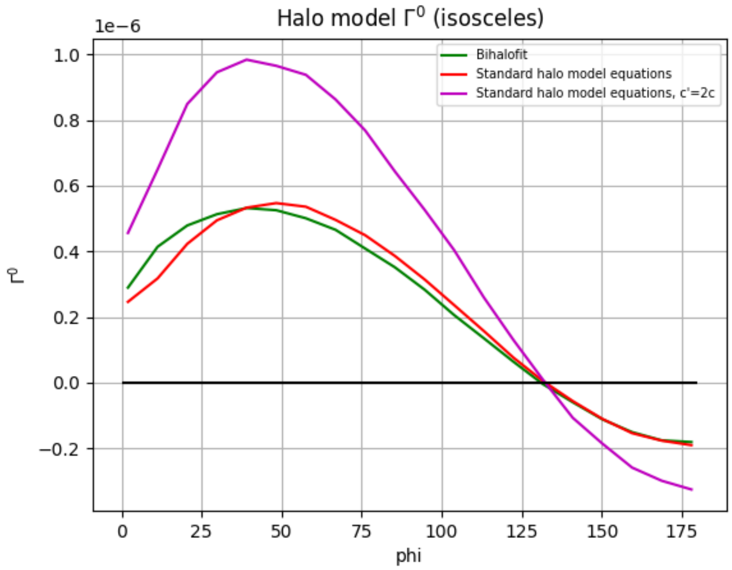

Figure 9. Isosceles Γ^0 shear 3pt function as a function of opening angle ϕ. Bihalofit compared with halo model integrals.)

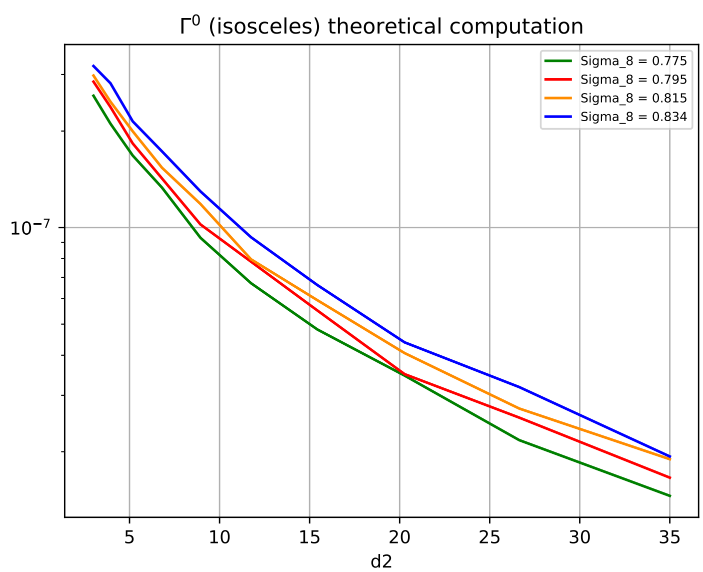

Figure 10. (Not yet updated) Isosceles Γ^0 shear 3pt function for varying σ_8. Number of evaluations: 25000